American Computer Science League

2021-2022 ● Contest 3: Short Problems ● Classroom Division

1. Boolean Algebra

Which ordered pair makes the following Boolean expression TRUE?

$$\overline{A}\,(AB + \overline{B})$$

A. (0, 0)
B. (0, 1)
C. (1, 0)
D. (1, 1)
E. None of the above

2. Boolean Algebra

Simplify the following Boolean expression:

$$A\,(\overline{B} + \overline{C}) + \overline{A}\,B + \overline{B\,C}$$

A. $\overline{A} + B + \overline{C}$
B. $B + \overline{A}\,C + \overline{C}$
C. $A\overline{B} + B + \overline{C}$
D. $\overline{B} + \overline{C} + \overline{A}\,B$
E. $C + \overline{A} + \overline{B}$

3. Data Structures

What nodes (listed left to right) are at a depth of 3 in the binary search tree for:

TOKYOOLYMPICS

A. S O I
B. C O L
C. I P S
D. K P Y
E. I O S

4. Data Structures

Given an initially empty queue and the following commands on the queue, what item will be popped next?

PUSH("A"), PUSH("Q"), PUSH("U"), X=POP( ),
PUSH("A"), PUSH("R"), X=POP( ), X=POP( ),
PUSH("I"), X=POP( ), PUSH("U"), PUSH("S")

A. A
B. I
C. R
D. U
E. S

5. FSAs and Regular Expressions

Which of the following strings match the regular expression pattern:

[ACSL]*[^aeiou]*21.2

a. Ab212
b. ACSL21
c. ACSL442122
d. cmptr212
e. A202122
f. pgrmmng21

A. a, c
B. b, c
C. c, e
D. d, e
E. e, f

6. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{A\,\overline{B}} + \overline{\overline{A}\,B\,\overline{C}} + \overline{\overline{B}\,C}$$

A. 0
B. 1
C. B
D. $\overline{A} + B + \overline{C}$
E. $\overline{A} + \overline{B} + \overline{C}$

7. Boolean Algebra

How many ordered triple(s) make the following Boolean expression TRUE?

$$A\,\overline{B} + \overline{A}\,(B + \overline{C})$$

A. 0
B. 3
C. 5
D. 7
E. 8

8. Data Structures

How many nodes have only a right child in the binary search tree for:

SUMMEROLYMPICS

A. 0
B. 1
C. 2
D. 3
E. 4

9. Data Structures

Given an initially empty stack and the following commands on the stack, what will be the next item popped?

PUSH("C"), PUSH("A"), PUSH("N"), X=POP( ),
X=POP( ), PUSH("I"), X=POP( ), PUSH("S"),
PUSH("M"), X=POP( ), X=POP( ), PUSH("A"),
PUSH("J"), X=POP( ), PUSH("O"), X=POP( ),
PUSH("R"), X=POP( )

A. S
B. M
C. I
D. C
E. A

10. FSAs and Regular Expressions

A new company is allowing its employees to select their own phone number and extension if it satisfies this regular expression:

1-321-[^05][0-9][^5]-[12][3-6][^7-9]8-[^0][2-6]

Which of the following are acceptable phone numbers and extensions?

a. 1-321-123-2348-11
b. 1-321-155-2368-96
c. 1-321-444-2448-44
d. 1-321-927-2408-84
e. 1-321-559-2328-13
f. 1-321-306-1366-62

A. a, f
B. b, c
C. e, f
D. c, d
E. a, e